

Abstraction and the Mathematical Basics

Sets, strings, relations, functions, trees, composition, and diagrams

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Mathematics begins by deciding what to ignore

Question

What remains the same across different linguistic events?

- An **abstraction** retains selected properties and suppresses others.
- An **invariant** is preserved across the variants admitted by that abstraction.
- Formal notation is useful only when it makes the selection explicit.

Take-away

The object of study is not the symbol. It is the structure the symbol allows us to compare.

MOLR Preface; §§1.1–1.3

Where we are

1. Sets and strings
2. Relations, functions, and trees
3. Composition and diagrams

A set records membership, not order or repetition

$$A = \{0, 2, 4, 126, 8, 1040, 0\} = \{0, 2, 4, 8, 126, 1040\}$$

Preserved

membership

Discarded

order and multiplicity

Guided analysis

Are $\{1\}$ and $\{1, 1, \dots, 1\}$ different sets? Explain without counting the written symbols.

MOLR §1.1

Set-builder notation states the admissibility condition

$$E = \{n \mid n \in \mathbb{N} \text{ and } n \text{ is even}\}$$

- The expression to the left of $|$ supplies a possible member.
- The condition to the right decides whether it is admitted.
- A finite list and a condition can describe the same set.

Guided analysis

Give two descriptions of the vowels in *linguistics*: one extensional, one intensional.

The empty set is not a set containing the empty set

$$\emptyset = \{\} \quad \text{but} \quad \{\emptyset\} \neq \emptyset$$

- $\emptyset$ has no members.
- $\{\emptyset\}$ has exactly one member.
- $\emptyset \subseteq A$ for every set A .

Question

What is $\mathcal{P}(\{a, b\})$?

Strings restore order and repetition

Let $\Sigma = \{a, b\}$. Then Σ^* contains

$\epsilon, a, b, aa, ab, ba, bb, \dots$

- ϵ is the empty *string*; $|\epsilon| = 0$.
- Concatenation is associative: $(uv)w = u(vw)$.
- It is not generally commutative: $uv \neq vu$.

Guided analysis

For $u = ab$ and $v = ba$, calculate uv , vu , and $|uv|$.

MOLR §1.2

Where we are

1. Sets and strings
2. Relations, functions, and trees
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A relation admits selected pairs

$$R \subseteq A \times B, \quad aRb \iff (a, b) \in R$$

Relation

CHILD-OF may relate one child to a parent and one parent to several children.

Properties

reflexive? symmetric?
antisymmetric? transitive?

Guided analysis

Which properties hold of ANCESTOR-OF? Which hold of SISTER-OF?

MOLR §1.3

A function is a relation with a stricter condition

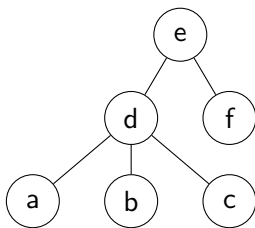
$f: A \rightarrow B$ assigns every $a \in A$ exactly one $b \in B$

- One input cannot receive two outputs.
- Different inputs may receive the same output.
- A function may be total on one chosen domain and partial on another.

Guided analysis

Is MOTHER-OF a function from people to people? Is CHILD-OF? State the assumptions your answer needs.

The same tree supports relations and functions



- CHILD-OF and SISTER-OF are relations.
- $p(x)$ = the parent of x is a function on nonroot nodes.
- A tree requires a unique root, unique parenthood, and connectedness.

Where we are

1. Sets and strings
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Composition connects matching domains

$$A \xrightarrow{g} B \xrightarrow{f} C \quad \Longrightarrow \quad f \circ g : A \rightarrow C$$
$$(f \circ g)(x) = f(g(x))$$

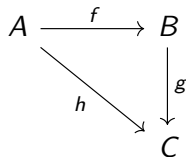
For $g(x) = x + 1$ and $f(y) = y^2$:

$$(f \circ g)(2) = 9, \quad (g \circ f)(2) = 5.$$

Take-away

Composition is associative, but composable functions need not commute.

A diagram asks whether two routes agree



commutes exactly when $h = g \circ f$.

- The diagram exposes alternative paths through a structure.
- Commutativity is a claim, not a property of every diagram.
- In linguistics it can test whether two analyses preserve the same relation.

MOLR §1.3, “Diagrams of functions”

Third hour: describe before calculating

For each item, state what the notation preserves and what it suppresses.

① $\{NP, VP, S\}$

② $NP \vee NP$

③ $R \subseteq A \times B$

④ $f: A \rightarrow B$

⑤ $g \circ f$

Guided analysis

Construct a three-node tree, write its parent function, and draw one commuting triangle based on it.

Week 1 synthesis

- ① Sets abstract away from order and repetition.
- ② Strings preserve both.
- ③ Relations admit pairs; functions admit exactly one output per input.
- ④ Trees combine relational structure with functional constraints.
- ⑤ Composition is possible only where domains connect.

Question

Next week: can function application, recursion, probability, and inference be seen as forms of connectedness?